Machine Intelligence 1

Practice Exam Questions

WiSe 23/24

Example Solution
4. September 2023

Important remarks:

- This is *not* a complete exam. It is only a collection of exam-like questions.
- The questions are meant as a preview and insight into what questions in the actual written exam *may* look like.
- It is representative for the style and the varying difficulty levels of the questions that could come up in the exam.
- It is *not* representative of
  - the duration of the exam. The duration of the actual written exam is 90 minutes.
  - the proportion of questions per topic. (a topic missing from here does not exclude the topic from the actual exam).
  - the proportion of question type (multiple choice vs. free text).
- *Do not overfit to these practice questions!*
- It is recommended to only look at these questions *after* you have studied the topics.
- The following rules apply in the actual written exam:
  - This is a closed-book exam. No cheat sheet, no books or any other resources are allowed.
  - You are only allowed to use pens and rulers, *nothing else*!
  - *No* calculators. You will not need one.
  - Do *not* use a pencil (Bleistift) or a pen that writes in red ink. Do *not* use whiteout (Tipp-Ex) or an ink eraser to delete wrong answers. Just cross out what you don't want to be graded. It is best to not bring any of these with you on the day of the exam.
  - Use only the paper included in the exam booklet to answer the questions. Do *not* bring any extra paper with you. Any answers on other sheets of paper are *invalid* and will be confiscated!
  - If you give an answer at any position other than directly following the question, mark clearly where the answer can be found (e.g. "The answer can be found on page XY").
  - Give brief but precise answers.
  - For derivations: show *how* you arrive at the solution.
  - An ambiguous answer is treated as a wrong answer (e.g. multiple answers to the same question or other ambiguities)

Question 1 (7 points) General.

Mark the correct answer to each of the the following questions. Mark it clearly. Exactly **one** answer is correct (earning at least 1 point). Wrong answers do not remove any points. Marking more than one answer to the same question invalidates all answers to that question. To remove an already marked answer, fill the entire square and draw an empty square next to it.

Example:

The answer is no longer → ☐ ■ Apple

But has been changed to → ☒ Banana

1. (1 point) To ensure *convergence* of online stochastic gradient descent, the *learning rate* η must...

   ☐ remain constant during training.
   ☒ shrink during training.
   ☐ grow during training.
   ☐ oscillate during training.

2. (2 points) Given is a regression task with p training samples $\underline{\mathbf{x}}^{(\alpha)} \in \mathbb{R}^N, \alpha \in \{1, \ldots, p\}$, and the corresponding labels $y_T^{(\alpha)} \in \mathbb{R}$. The model class is a linear neuron without bias, $y(\underline{\mathbf{x}}; \underline{\mathbf{w}}) = \underline{\mathbf{w}}^\top \underline{\mathbf{x}}$, and the regularized cost function R^T is:

$$R^T(\underline{\mathbf{w}}) \quad = \quad \frac{1}{2p} \sum_{\alpha=1}^{p} \Big(y(\underline{\mathbf{x}}^{(\alpha)}; \underline{\mathbf{w}}) - y_T^{(\alpha)} \Big)^2 \quad + \quad \frac{\lambda}{2p} \big\| \underline{\mathbf{w}} \big\|^2 .$$

   Which of the following statements is <u>*correct*</u>?

   ☐ The optimization problem cannot be solved analytically when expanding the feature vectors $\underline{\mathbf{x}}^{(\alpha)}$ of the data set using nonlinear (e.g. monomial) basis functions $\{\phi_1, \phi_2, \ldots \phi_K\}$, which corresponds to changing the data set to $\{(\underline{\phi}^{(\alpha)}, y_T^{(\alpha)})\}_{\alpha=1}^p$, where $\underline{\phi}^{(\alpha)} = (\phi_1(\underline{\mathbf{x}}^{(\alpha)}),\ \phi_2(\underline{\mathbf{x}}^{(\alpha)}),\ \ldots,\ \phi_K(\underline{\mathbf{x}}^{(\alpha)}))^\top$.
   ☐ A suitable alternative training error measure is the cross entropy cost function.
   ☐ Replacing the regularization term in R^T with an L1-regularization typically leads to more nonzero components in the optimal weight vector.
   ☒ The optimal weight vector is given by

$$\underline{\mathbf{w}}^* \quad = \quad \Big(\sum_{\alpha=1}^{p} \underline{\mathbf{x}}^{(\alpha)} \big(\underline{\mathbf{x}}^{(\alpha)}\big)^\top + \lambda \underline{\mathbf{I}} \Big)^{-1} \Big(\sum_{\alpha=1}^{p} \underline{\mathbf{x}}^{(\alpha)} y_T^{(\alpha)} \Big),$$

   where $\underline{\mathbf{I}} \in \mathbb{R}^{N,N}$ is the identity matrix.

3. (1 point) It was observed that the training error of a model was smaller than its generalization error. From this we conclude that...

   ☐ our model is overfitting.
   ☐ we need more input features.
   ☐ we need to transform the training data into a higher dimensional space.
   ☒ nothing, not enough information provided.

4. (2 points) Which of the following statements is *correct* in the context of a binary classification problem? Here the data set is $\underline{\mathbf{x}}^{(\alpha)} \in \mathbb{R}^N, \alpha \in \{1, \ldots, p\}$ with corresponding labels $y_T^{(\alpha)} \in \{0, 1\}$. The model $y(\underline{\mathbf{x}}; \underline{\mathbf{w}})$ outputs values in the range $(0, 1)$ representing the probability that $\underline{\mathbf{x}}$ has the class label $y_T = 1$.

   ☒ A suitable cost function is given by the cross entropy, $E^T = \frac{1}{p} \sum_{\alpha=1}^{p} \left\{ -y_T^{(\alpha)} \ln y(\underline{\mathbf{x}}^{(\alpha)}) - (1 - y_T^{(\alpha)}) \ln \left[1 - y(\underline{\mathbf{x}}^{(\alpha)}) \right] \right\}$

   ☐ The classification accuracy, i.e., the fraction of correctly classified data points, can be optimized directly by gradient descent.

   ☐ Overfitting is a problem only for regression tasks not for classification.

   ☐ The class label predicted by the model for a data point $\underline{\mathbf{x}}$ is given by $y(\underline{\mathbf{x}}; \underline{\mathbf{w}}) = 1$ if $y(\underline{\mathbf{x}}; \underline{\mathbf{w}}) \leq \frac{1}{2}$ and $y(\underline{\mathbf{x}}; \underline{\mathbf{w}}) = 0$ otherwise.

5. (1 point) Assume a radial basis function network makes predictions by $y(\underline{\mathbf{x}}; \underline{\mathbf{w}}) = \sum_{i=1}^{K} w_i \exp\left(-\frac{1}{2\sigma_i^2} \|\underline{\mathbf{x}} - \underline{\mathbf{t}}_i\|^2 \right)$. Which of the following statement is *wrong* in the context of training the network?

   ☐ The weights w_i can be calculated analytically.

   ☐ The variances σ_i^2 can be heuristically determined.

   ☐ The centers $\underline{\mathbf{t}}_i$ can be obtained by K-means clustering.

   ☒ Extending a data set to include additional unlabeled data points has no effect on the performance of the network.

Question 2 (5 points) Neural Networks.

Mark the correct answer to each of the the following questions. Mark it clearly. Exactly **one** answer is correct (earning at least 1 point). Wrong answers do not remove any points. Marking more than one answer to the same question invalidates all answers to that question. To remove an already marked answer, fill the entire square and draw an empty square next to it.

Example:

The answer is no longer → ☐ ■ Apple

But has been changed to → ☒ Banana

1. (1 point) Which of the following statements about multi-layer perceptrons (MLP) is *true*?

   ☐ MLPs can only have one output neuron.
   ☒ MLPs cannot contain cycles.
   ☐ Neurons in an MLP can only be connected to the previous layer.
   ☐ Each layer of an MLP must be fully connected to the next layer.

2. (1 point) The weights that globally minimize the training error of an MLP...

   ☐ are determined by descending the gradient of the training error for a single random weights initialization.
   ☐ are unique.
   ☒ can correspond to an overfitted neural network.
   ☐ represents a bad fit for the training data.

3. (2 points) Which statement is *wrong* when solving a multi-class classification problem with one-hot encoded labels $\underline{\mathbf{y}}_T^{(\alpha)} \in \{0,1\}^M$, where M is the number of classes, using an MLP in the usual way?

   ☒ The output layer consists of M neurons with the logistic transfer function for each of them.
   ☐ The k-th component $y_k(\underline{\mathbf{x}}^{(\alpha)}; \underline{\mathbf{w}})$ of the network output $\underline{\mathbf{y}}(\underline{\mathbf{x}}; \underline{\mathbf{w}})$ represents the probability of $\underline{\mathbf{x}}$ belonging to class k.
   ☐ The cost function for training can be defined by $e^{(\alpha)} = -\sum_{k=1}^{M} y_T^{(\alpha)} \ln y_k(\underline{\mathbf{x}}^{(\alpha)}; \underline{\mathbf{w}})$.
   ☐ The normalization of the labels $\sum_{k=1}^{M} y_{T,k}^{(\alpha)} = 1$ is reflected in the model by $y_k(\underline{\mathbf{x}}^{(\alpha)}; \underline{\mathbf{w}}) = 1 - \sum_{j=1, j\neq k}^{M} y_j(\underline{\mathbf{x}}^{(\alpha)}; \underline{\mathbf{w}})$.

4. (1 point) Which statement is *wrong* for a convolutional neural network?

   ☐ The weights of a convolution filter are shared between all neurons of the same feature map.
   ☒ A different number of feature maps for the same layer leads to the same model complexity.
   ☐ A convolutional layer without pooling corresponds to a fully connected layer with a sparse connection matrix that has identical but shifted entries in each inner row.
   ☐ The max-pooling operation down-samples information and leads to invariance to subtle translations.

Question 3 (7 points) Neural Networks.

Assume you want to solve a regression problem for a data set $\{(\underline{\mathbf{x}}^{(\alpha)}, y_T^{(\alpha)})\}_{\alpha=1}^{p}$, with observations $\underline{\mathbf{x}}^{(\alpha)} \in \mathbb{R}^N$ and labels $y_T^{(\alpha)} \in \mathbb{R}$, using a multi-layer perceptron parametrized by $\underline{\mathbf{w}}$ (containing all weights and biases). The network has one neuron in the output layer yielding the scalar model prediction $y(\underline{\mathbf{x}}; \underline{\mathbf{w}})$ for any $\underline{\mathbf{x}} \in \mathbb{R}^N$.

1. (1 point) Let $e^{(\alpha)}(\underline{\mathbf{w}})$ be the error of predicting the label $y_T^{(\alpha)}$ with the model output $y(\underline{\mathbf{x}}^{(\alpha)}; \underline{\mathbf{w}})$. Write down the average error E^T of the data set without assuming a specific cost function.

   Solution:

   $$E^T(\underline{\mathbf{w}}) = \frac{1}{p} \sum_{\alpha=1}^{p} e^{(\alpha)}(\underline{\mathbf{w}})$$

2. (4 points) Assume that you want to improve the model using a single gradient descent step that updates the weight vector $\underline{\mathbf{w}}_{(t)}$ of the previous step according to

   $$\underline{\mathbf{w}}_{(t+1)} = \underline{\mathbf{w}}_{(t)} + \eta_t \, \Delta\underline{\mathbf{w}}_{(t)}. \qquad (*)$$

   Specify the weight update for the following three alternatives of gradient descent in terms of $e^{(\alpha)}(\underline{\mathbf{w}})$ only:

   - batch algorithm:

     Solution:
     1P
     $$\Delta\underline{\mathbf{w}}_{(t)} = -\frac{1}{p} \sum_{\alpha=1}^{p} \frac{\partial e^{(\alpha)}}{\partial \underline{\mathbf{w}}}(\underline{\mathbf{w}}_{(t)})$$

   - online algorithm:

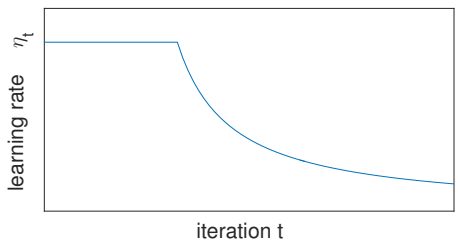
     Solution:
     1P
     $$\Delta\underline{\mathbf{w}}_{(t)} = -\frac{\partial e^{(\alpha)}}{\partial \underline{\mathbf{w}}}(\underline{\mathbf{w}}_{(t)}) \quad \text{for random data point } \alpha$$

   - mini-batch algorithm:

     Solution:
     2P
     $$\Delta\underline{\mathbf{w}}_{(t)} = -\frac{1}{|\mathcal{I}_{(t)}|} \sum_{\alpha \in \mathcal{I}_{(t)}} \frac{\partial e^{(\alpha)}}{\partial \underline{\mathbf{w}}}(\underline{\mathbf{w}}_{(t)}) \quad \text{for random index set } \mathcal{I}_{(t)} \subset \{1, \ldots, p\} \text{ of the mini-batch size}$$

3. (2 points) Plot a learning rate schedule ($\eta_{(t)}$ vs. iterations t) for which the parameters $\underline{\mathbf{w}}(t)$ (cf. equation (*) above) converge to a stationary point of E^T when $t \to \infty$. Make clear that the plotted learning rate satisfies the Robbins-Monro conditions by specifying mathematically a suitable $\eta_{(t)}$ decay behaviour into the respective part of the plot.

   Solution:
   1.5P

   0.5P
   $\eta_{(t)} \propto \dfrac{1}{t - t_0 + 1}$ for $t > t_0$ with end of plateau t_0;
   also acceptable: $\eta_{(t)} \propto 1/t$ (with and without proportionality).

Question 4 (3 points) Recurrent Neural Networks.

Mark the correct answer to each of the the following questions. Mark it clearly. Exactly **one** answer is correct (earning at least 1 point). Wrong answers do not remove any points. Marking more than one answer to the same question invalidates all answers to that question. To remove an already marked answer, fill the entire square and draw an empty square next to it.

Example:

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But has been changed to → ☒ Banana

1. (3 points) Which of the following statements for a simple recurrent neural network is *wrong*?
   Let $\left\{\left(\left(\underline{\mathbf{x}}^{(\alpha,t)}, y_T^{(\alpha,t)}\right)\right)_{t=1}^{n_\alpha}\right\}_{\alpha=1}^{p}$ be the data set with $\underline{\mathbf{x}}^{(\alpha,t)} \in \mathbb{R}^N$ and $y_T^{(\alpha,t)} \in \{0,1\}$.

   ☒ The computational complexity of backpropagation through time is $O(n_\alpha^2)$ for the time series α.

   ☐ The training cost is defined by $E^T = \frac{1}{p}\sum_{\alpha=1}^{p}\frac{1}{n_\alpha}\sum_{t=1}^{n_\alpha} e^{(\alpha,t)}$ where $e^{(\alpha,t)}$ is the error of an individual time point.

   ☐ The hidden neurons are jointly defined via $\underline{\mathbf{s}}^{(t)} = \underline{\mathbf{f}}\big(\underline{\mathbf{U}}\,\underline{\mathbf{x}}^{(t)} + \underline{\mathbf{W}}\,\underline{\mathbf{s}}^{(t-1)} + \underline{\mathbf{b}}^{\mathrm{s}}\big)$ with $\underline{\mathbf{f}} = (f,\ldots,f)^\top$ being the (componentwise applied) transfer transfer function f (e.g., $f = \tanh$), $\underline{\mathbf{U}}$ and $\underline{\mathbf{W}}$ the input-to-hidden and hidden-to-hidden weight matrices, $\underline{\mathbf{x}}^{(t)}$ the input at time t, and $\underline{\mathbf{b}}^{\mathrm{s}}$ the bias vector.

   ☐ The output neurons are jointly defined through $\underline{\mathbf{y}}^{(t)} = \underline{\mathbf{g}}\big(\underline{\mathbf{V}}\,\underline{\mathbf{s}}^{(t)} + \underline{\mathbf{b}}^{\mathrm{y}}\big)$, where $\underline{\mathbf{g}} = (g,\ldots,g)^\top$ being the (componentwise applied) output transfer function g (e.g. logistic function) and $\underline{\mathbf{V}}$ the hidden-to-output weight matrices, $\underline{\mathbf{s}}(t)$ is the output of the hidden neurons at time t, and $\underline{\mathbf{b}}^{\mathrm{y}}$ the bias vector.

Question 5 (3 points) Support Vector Machine (SVM).

Mark the correct answer to each of the the following questions. Mark it clearly. Exactly **one** answer is correct (earning at least 1 point). Wrong answers do not remove any points. Marking more than one answer to the same question invalidates all answers to that question. To remove an already marked answer, fill the entire square and draw an empty square next to it.

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1. (1 point) Consider the solution found from training a maximum margin classifier. Under what condition is the solution unique?

   ☐ The data is perfectly separable.
   ☐ The data is linearly separable.
   ☐ The solution is only unique if we sphere the data beforehand.
   ☒ Always.

2. (1 point) Which of the following statements is *wrong*?

   ☐ Slack variables of C-SVM allow some data points to be misclassified.
   ☐ The C-SVM performs a linear classification in (often high-dimensional) feature space when using a kernel.
   ☒ An individual C-SVM model can solve multi-class classification problems with one-hot encoded labels.
   ☐ The C-SVM does not allow for margin errors in the training data set when $C \to \infty$.

3. (1 point) Which statement is *wrong* for the parameter C of a C-SVM?

   ☐ Its value is determined by hyperparameter optimization, e.g., through n-fold cross-validation.
   ☐ A large value can lead to overfitting to the training data.
   ☒ C occurs in the inequality constraints of the primal optimization problem.
   ☐ C occurs in the inequality constraints of the dual optimization problem.

Question 6 (5 points) C-SVMs.

A C-SVM classifier is trained on a dataset containing p samples $\underline{\mathbf{x}}^{(\alpha)} \in \mathbb{R}^N, \alpha \in \{1, \dots, p\}$, with the corresponding labels $y_T^{(\alpha)} \in \{-1, +1\}$.

1. (3 points) Define the primal problem for the C-SVM and name the introduced variables. (Definition of the Lagrangian and the dual problem is **not** necessary to answer this question.)

   Solution:
   The primal problem of the C-SVM, with *slack-variables* $\varphi_\alpha, \forall \alpha \in \{1, \dots, p\}$, is:

   $$\min_{\underline{\mathbf{w}}} \tfrac{1}{2}\|\underline{\mathbf{w}}\|^2 + \tfrac{C}{p}\sum_{\alpha=1}^{p}\varphi_\alpha\,, \qquad \text{s.t.} \quad y_T^{(\alpha)}\big(\underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} + b\big) \geq 1 - \varphi_\alpha\,, \quad \forall \alpha \in \{1, \dots, p\}\,, \quad \varphi_\alpha \geq 0\,.$$

   The variables φ_α are called "slack-variables" and C is a regularization parameter.
   C instead of $\frac{C}{p}$ is also correct. Using other letters is also ok, as long as it's clear what they stand for.

2. (2 points) Describe in not more than 3 short sentences how the *kernel-trick* can be used to perform classification in high-dimensional feature spaces?

   Solution:
   By replacing all inner products $(\underline{\mathbf{x}}^{(\alpha)})^\top \underline{\mathbf{x}}^{(\beta)}$ by a kernel $k(\underline{\mathbf{x}}^{(\alpha)}, \underline{\mathbf{x}}^{(\beta)})$, the algorithm performs classification in a high-dimensional feature space, for which the kernel is an inner product.

Question 7 (5 points) Bayesian Networks and Graphical Models.

Mark the correct answer to each of the the following questions. Mark it clearly. Exactly **one** answer is correct (earning at least 1 point). Wrong answers do not remove any points. Marking more than one answer to the same question invalidates all answers to that question. To remove an already marked answer, fill the entire square and draw an empty square next to it.

Example:

The answer is no longer → ☐ ■ Apple

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1. (1 point) For a set of query variables X, observed variables E, and unobserved variables Y, *inference* computes:

   ☐ $P(E|Y)$
   ☐ $P(E|X,Y)$
   ☒ $P(X|E)$
   ☐ $P(X|E,Y)$

2. (2 points) Which of the following statements is <u>*true*</u>.

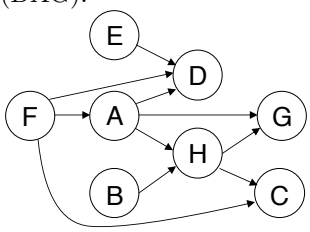
   ☐ The message passed from the root clique ABC, having a potential function $f_1(A,B,C)$, to the separator BC can be computed as $\mu_{ABC\to BC} = \sum_{B,C} f_1(A,B,C)$.
   ☒ The conditional distributions that underly a directed acyclic graph are used to initialize the clique potentials of a junction tree for message passing.
   ☐ All nodes on the path between two cliques in a junction tree, which both contain variable X may not contain X.
   ☐ For each moral graph exactly one chordal graph exists.

3. (2 points) Which of the following statements is <u>*wrong*</u>?

   ☐ Evidence, that a random variable X has value x, is introduced into a junction tree by adding an additional node with potential function $f_{\text{evidence}}(X) = \begin{cases} 1 \text{ for } X = x \\ 0 \text{ for } X \neq x. \end{cases}$
   ☐ Some marginal probability distributions are available before completing the distribute pass of the message passing on a junction tree.
   ☐ Passing a message along one single edge of a junction tree involves summing over at most as many variables as there are elements in the largest clique.
   ☒ Every junction tree corresponds to a unique directed acyclic graph.

Question 8 (9 points) Bayesian Networks (continued).

Given is the following directed acyclic graph (DAG).

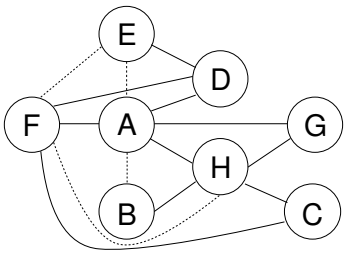


1. (1 point) Write down the factorization of the joint distribution that follows from the DAG.

   Solution:
   $P(A, B, C, D, E, F, G, H) = P(F)\, P(E)\, P(A|F)\, P(B)\, P(D|A, E, F)\, P(H|A, B)\, P(G|A, H)\, P(C|F, H)\,.$

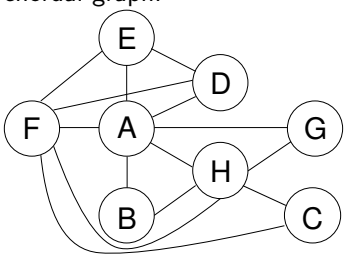
2. (2 points) Convert the DAG into the corresponding moral graph.

   Solution:
   $\frac{1}{2}$ point for each of the added undirected edges: EF, EA, AB and FH



3. (2 points) Construct a chordal graph from the moral graph.

   Solution:
   There are no chordless cycles in the moral graph. No triangulation required. This moral graph is also a chordal graph.

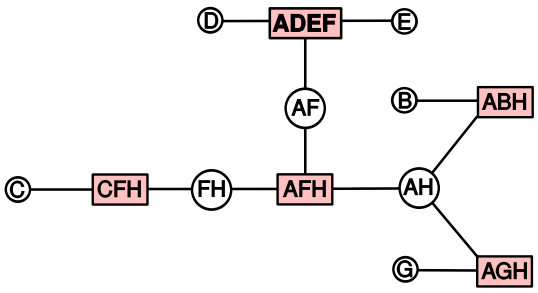


4. (2 points) Identify all cliques of the chordal graph. Write them down (ordered alphabetically) in the format: VXZ, VYZ, WXZ, ...

   Solution:
   The cliques are: ABH, ADEF, AFH, AGH, CFH

5. (2 points) Construct a junction tree from the cliques and seperators of the chordal graph.

   Solution:
   The leaf nodes are optional but not required.



Question 9 (5 points) Reinforcement learning.

Mark the correct answer to each of the the following questions. Mark it clearly. Exactly **one** answer is correct (earning at least 1 point). Wrong answers do not remove any points. Marking more than one answer to the same question invalidates all answers to that question. To remove an already marked answer, fill the entire square and draw an empty square next to it.

1. (1 point) Which of the following statements about reinforcement learning is *false*?

   ☐ The value function is defined as expected infinite sum of discounted future rewards.

   ☐ States must describe all relevant variables of the environment.

   ☐ A probabilistic transition model can describe deterministic environments.

   ☒ The policy is part of the the environment.

2. (2 points) Which of the following equations does *not* correspond to the Bellman equation?
   Note that the action is drawn from the policy π, i.e., $\underline{\mathbf{a}}^{(0)} \sim \pi(\cdot|\underline{\mathbf{x}}^{(0)})$, and the next state is drawn from the transition model P, i.e., $\underline{\mathbf{x}}^{(1)} \sim P(\cdot|\underline{\mathbf{x}}^{(0)}, \underline{\mathbf{a}}^{(0)})$.

   ☐ $V^\pi(\underline{\mathbf{x}}_i) = \sum_{k=1}^{A} \pi(\underline{\mathbf{a}}_k \,|\, \underline{\mathbf{x}}_i)\Big(r(\underline{\mathbf{x}}_i, \underline{\mathbf{a}}_k) + \gamma \sum_{j=1}^{S} P(\underline{\mathbf{x}}_j \,|\, \underline{\mathbf{x}}_i, \underline{\mathbf{a}}_k)\, V^\pi(\underline{\mathbf{x}}_j)\Big)$

   ☐ $V^\pi(\underline{\mathbf{x}}_i) = \mathbb{E}_{\underline{\mathbf{a}}^{(0)}, \underline{\mathbf{x}}^{(1)}}\left[r\left(\underline{\mathbf{x}}^{(0)}, \underline{\mathbf{a}}^{(0)}\right) + \gamma\, V^\pi\left(\underline{\mathbf{x}}^{(1)}\right) \,\middle|\, \underline{\mathbf{x}}^{(0)} = \underline{\mathbf{x}}_i\right]$

   ☒ $V^\pi(\underline{\mathbf{x}}_i) = r\left(\underline{\mathbf{x}}_i, \underline{\mathbf{a}}^{(0)}\right) + \mathbb{E}_{\underline{\mathbf{a}}^{(0)}, \underline{\mathbf{x}}^{(1)}}\left[\gamma\, V^\pi\left(\underline{\mathbf{x}}^{(1)}\right) \,\middle|\, \underline{\mathbf{x}}^{(0)} = \underline{\mathbf{x}}_i\right]$

   ☐ $V^\pi(\underline{\mathbf{x}}_i) = \mathbb{E}_{\underline{\mathbf{a}}^{(0)}}\left[r\left(\underline{\mathbf{x}}_i, \underline{\mathbf{a}}^{(0)}\right)\right] + \mathbb{E}_{\underline{\mathbf{a}}^{(0)}, \underline{\mathbf{x}}^{(1)}}\left[\gamma\, V^\pi\left(\underline{\mathbf{x}}^{(1)}\right) \,\middle|\, \underline{\mathbf{x}}^{(0)} = \underline{\mathbf{x}}_i\right]$

3. (1 point) Which of the following statements is *wrong*:

   ☒ A large value for the discount factor leads to fast converging value iteration.

   ☐ A large value for the discount factor makes rewards far in the future more relevant.

   ☐ The runtime complexity of the analytical value function method is the same for a small discount factor value and for a large value of γ.

   ☐ The value function V^π is parametrized by the discount factor γ.

4. (1 point) $Q^{\pi^*}(\underline{\mathbf{x}}_i, \underline{\mathbf{a}}_k)$ and $V^{\pi^*}(\underline{\mathbf{x}}_i)$ represent the optimal state-action and optimal value function respectively. Which of the following extracts the optimal policy $\pi^*(\underline{\mathbf{a}}_k|\underline{\mathbf{x}}_i)$?

   ☐ $\pi^*(\underline{\mathbf{a}}_k|\underline{\mathbf{x}}_i) = \sum_{k=1}^{A} Q^{\pi^*}(\underline{\mathbf{x}}_i, \underline{\mathbf{a}}_k)$

   ☐ $\pi^*(\underline{\mathbf{a}}_k|\underline{\mathbf{x}}_i) = \mathbb{E}\left[\, V^{\pi^*}(\underline{\mathbf{x}}_i)\,\right]$

   ☐ $\pi^*(\underline{\mathbf{a}}_l|\underline{\mathbf{x}}_i) = \begin{cases} 1, & \text{if } l = \operatorname*{argmax}_{1 \le l \le S} V^{\pi^*}(\underline{\mathbf{x}}_l) \\ 0, & \text{else.} \end{cases}$

   ☒ $\pi^*(\underline{\mathbf{a}}_k|\underline{\mathbf{x}}_i) = \begin{cases} 1, & \text{if } k = \operatorname*{argmax}_{1 \le k \le A} Q^{\pi^*}(\underline{\mathbf{x}}_i, \underline{\mathbf{a}}_k) \\ 0, & \text{else.} \end{cases}$

Question 10 (7 points) Reinforcement learning: Evaluation.

A Markov Decision Process consists of a set of states $\underline{\mathbf{x}} \in \mathcal{X} := \{\underline{\mathbf{x}}_1, \dots, \underline{\mathbf{x}}_S\}$, a set of actions $\underline{\mathbf{a}} \in \mathcal{A} := \{\underline{\mathbf{a}}_1, \dots, \underline{\mathbf{a}}_A\}$, a transition model $P(\underline{\mathbf{x}}_j | \underline{\mathbf{x}}_i, \underline{\mathbf{a}}_k)$ and a bounded reward function $r(\underline{\mathbf{x}}_i, \underline{\mathbf{a}}_k)$.
The value function $V^\pi(\underline{\mathbf{x}}_i)$ for a fixed policy π is defined by

$$V^\pi(\underline{\mathbf{x}}_i) = \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t \, r(\underline{\mathbf{x}}^{(t)}, \underline{\mathbf{a}}^{(t)}) \,\middle|\, \underline{\mathbf{x}}^{(0)} := \underline{\mathbf{x}}_i \right],$$

$$\text{where } \underline{\mathbf{a}}^{(t)} \sim \pi(\cdot \,|\, \underline{\mathbf{x}}^{(t)}) \;,\; \underline{\mathbf{x}}^{(t+1)} \sim P(\cdot \,|\, \underline{\mathbf{x}}^{(t)}, \underline{\mathbf{a}}^{(t)}) \text{ and } \gamma \in [0, 1).$$

1. (1 point) The Bellman equation is defined as

$$\underline{\mathbf{v}}^\pi = \underline{\mathbf{r}}^\pi + \gamma \underline{\mathbf{P}}^\pi \underline{\mathbf{v}}^\pi \,, \quad \text{with} \begin{cases} \mathrm{v}_i^\pi &:= V^\pi(\underline{\mathbf{x}}_i) \\ r_i^\pi &:= \sum_{k=1}^{A} \pi(\underline{\mathbf{a}}_k \,|\, \underline{\mathbf{x}}_i) \, r(\underline{\mathbf{x}}_i, \underline{\mathbf{a}}_k) \\ P_{ij}^\pi &:= \sum_{k=1}^{A} \pi(\underline{\mathbf{a}}_k \,|\, \underline{\mathbf{x}}_i) \, P(\underline{\mathbf{x}}_j | \underline{\mathbf{x}}_i, \underline{\mathbf{a}}_k) \end{cases}$$

   It decomposes the value function into two parts, namely...

   Solution:
   ...the expected immediate reward and the discounted value of the next state.

2. (6 points) Derive the Bellman equation from the value function $V^\pi(\underline{\mathbf{x}}_i)$.
   Hint: Start with writing down the first few terms of the sum.
   Also: Use a "short-hand" notation (e.g. $r^0 := r(\underline{\mathbf{x}}^{(0)}, \underline{\mathbf{a}}^{(0)})$).

   Solution:

$$\begin{aligned}
V^\pi(\underline{\mathbf{x}}_i) &= \mathbb{E}\Big[\textstyle\sum_{t=0}^{\infty} \gamma^t \, r(\underline{\mathbf{x}}^{(t)}, \underline{\mathbf{a}}^{(t)}) \,\Big|\, \underline{\mathbf{x}}^{(0)} := \underline{\mathbf{x}}_i \Big] \\
&= \mathbb{E}\Big[\textstyle\sum_{t=0}^{\infty} \gamma^t \, r^{(t)} \,\Big|\, \underline{\mathbf{x}}^{(0)} := \underline{\mathbf{x}}_i \Big] \\
&= \mathbb{E}\Big[r^{(0)} \,+\, \gamma \, r^{(1)} \,+\, \gamma^2 \, r^{(2)} + \dots \,\Big|\, \underline{\mathbf{x}}^{(0)} := \underline{\mathbf{x}}_i \Big] \\
&= \mathbb{E}\Big[r^{(0)} \Big] + \mathbb{E}\Big[\gamma \, r^{(1)} \,+\, \gamma^2 \, r^{(2)} + \dots \,\Big|\, \underline{\mathbf{x}}^{(0)} := \underline{\mathbf{x}}_i \Big] \\
&= \mathbb{E}\Big[r^{(0)} \Big] + \gamma \, \mathbb{E}\Big[r^{(1)} \,+\, \gamma \, r^{(2)} + \dots \,\Big|\, \underline{\mathbf{x}}^{(0)} := \underline{\mathbf{x}}_i \Big] \\
&= \mathbb{E}\Big[r^{(0)} \Big] + \gamma \, \mathbb{E}\Big[V^\pi\left(\underline{\mathbf{x}}^{(1)}\right) \,\Big|\, \underline{\mathbf{x}}^{(0)} := \underline{\mathbf{x}}_i \Big] \\
&= \quad r_i^\pi \quad + \, \gamma \, P_{ij}^\pi \, V^\pi\left(\underline{\mathbf{x}}_j\right)
\end{aligned}$$